HBRP PUBLICATION

Journal of Advanced Research in Artificial Intelligence & It's Applications

e-ISSN: *3048-6440*

Volume 03 Issue 01
Jan-Apr, 2026

***Corresponding Author:** **Sona Jaison,** Department of Computer Science Engineering, St. Joseph's College of Engineering and Technology
Kottayam, Kerala

Insightify: An AI Based Mock Interview Platform

Beegum Aliya Arif[1], Chinju Biju[2], Mariya Michael[3], Sona Jaison[4], Anu V Kottath[5]

[1-5]Department of Computer Science Engineering, St. Joseph's College of Engineering and Technology Kottayam, Kerala

ABSTRACT

In the fast-evolving landscape of technology and competitive job markets, Artificial Intelligence (AI) is transforming the way candidates prepare for interviews. AI-driven mock interview systems leverage advanced technologies such as Natural Language Processing (NLP), speech recognition, and text-to- speech capabilities to create realistic interview simulations. These intelligent platforms provide an interactive and adaptable practice environment, enabling job seekers to refine their responses effectively.

A major advantage of AI-based mock interviews is their ability to assess both verbal and non-verbal aspects of communication. By analyzing speech patterns, tone, clarity, fluency, and confidence levels, these systems offer detailed, real-time feedback that helps candidates improve their communication skills. Further- more, AI models can be customized to simulate different job roles, industries, and difficulty levels, making interview practice more personalized and relevant.

This paper explores the core technologies behind AI-powered mock interviews, highlighting their advantages over traditional preparation methods. Additionally, we discuss challenges such as ethical considerations, data privacy, and the need for continuous algorithm updates to ensure fairness and accuracy. Through case studies and data-driven insights, we examine how AI is bridging the gap between theoretical learning and real-world application, ultimately boosting candidates' confidence and employability.

Keywords:*-Artificial Intelligence, Mock Interviews, Natural Language Processing, Speech Recognition, Job Market, Machine Learning, Career Preparation.*

1. INTRODUCTION

The rapid evolution of technology and the ever-changing job market dynamics have created a need for more sophisticated and effective interview preparation tools. Traditional methods of preparing for job interviews, such as face-to-face mock interviews or written practice questions, often fail to address the complexities of modern recruitment processes [1].

These approaches are time-consuming, resource-intensive, and unable to provide immediate and personalized feedback to candidates [2]. Furthermore, the reliance on human interview- ers introduces variability in assessments, making it difficult to maintain consistency and fairness across different candidates [3].

Artificial Intelligence (AI) has emerged as a transformative solution to these challenges, offering a highly scalable and intelligent approach to interview preparation [4]. AI-powered mock interview systems integrate state-of-the-art technologies such as Natural Language Processing (NLP), machine learning, speech recognition, and sentiment analysis to create an interactive and immersive experience [5]. These systems simulate real interview environments by assessing candidates' responses in real-time, analyzing their tone, fluency, confidence, and articulation, and providing instant feedback to help refine their communication and presentation skills [6].

The integration of AI-driven assessment mechanisms allows candidates to engage in a data-driven and iterative learning process [7]. Unlike traditional methods that rely on static feedback, AI-based systems continuously evolve based on user interactions, adapting to various industries, job roles, and difficulty levels [8]. This personalized approach ensures that candidates receive targeted training specific to their desired career paths, making interview preparation more relevant and impactful [9].

Moreover, AI-based mock interviews go beyond verbal responses by incorporating facial recognition and body language analysis to assess non-verbal communication skills [10]. These advanced capabilities enable candidates to develop a holistic understanding of their interview performance, addressing both verbal and non-verbal aspects crucial for making a positive impression on recruiters [11].

The significance of AI-powered mock interview platforms lies in their ability to bridge the gap between theoretical knowledge and real-world application [12]. These systems not only help candidates improve their technical responses but also enhance their soft skills, including confidence, active listening, and adaptability—qualities that are highly valued by employers [13]. Additionally, the ability to conduct interviews in multiple languages broadens accessibility, catering to a diverse pool of job seekers worldwide [14].

Another advantage of AI-based interview systems is their potential to reduce anxiety and build confidence in candidates [15]. Many job seekers experience nervousness and hesitation before interviews, which can affect their performance. AI- powered platforms allow users to practice repeatedly in a low- pressure environment, helping them become more comfortable with answering questions, structuring their responses, and handling challenging situations effectively [16]. By simulating high-pressure interview scenarios, these tools prepare candidates to remain composed and articulate in actual interviews [17].

Furthermore, AI-driven mock interview systems help standardize the interview preparation process. Unlike traditional mock interviews, which may be influenced

by interviewer biases, AI-powered platforms rely on data-driven evaluation metrics to ensure fairness and consistency in assessments [18]. This standardization promotes an equitable opportunity for all candidates, regardless of their background, making the job search process more inclusive and transparent.

Despite their numerous advantages, AI-based interview systems also present certain challenges. The accuracy of assessments depends on the quality of AI models, and there is a need for continuous improvements to minimize biases in evaluation [19]. Additionally, data privacy and security concerns must be addressed to protect user information and maintain trust in the system [20]. Ethical considerations, such as ensuring AI does not disadvantage certain groups due to inherent biases in training data, must also be carefully managed.

This paper explores the foundational technologies of AI- driven mock interview systems, their advantages over traditional interview preparation methods, and their overall impact on job seekers. We also discuss critical challenges such as data privacy concerns, biases in AI-driven evaluations, and the need for ongoing refinements to ensure ethical and unbiased assessments. Through detailed analysis and case studies, we highlight how AI is redefining the future of interview preparation, making it an indispensable tool for job seekers striving for career success.

2. LITERATURE SURVEY

AI-driven mock interview systems have undergone substantial advancements, transitioning from traditional rule-based models to more adaptive and intelligent solutions. Early systems, such as the chatbot-based simulation developed by Smith et al. [21], relied on pre-scripted interactions, pro- viding structured practice but failing to adapt dynamically to candidate responses. Similarly, HireVue introduced AI- driven video analytics, assessing facial expressions, speech tone, and emotional cues [22], but it primarily focused on non-verbal elements rather than engaging in deep natural lan- guage processing (NLP) to evaluate content quality. OpenAI Codex [23] advanced the field by integrating NLP-driven question generation and evaluation, enabling a more interactive experience where AI provided real-time feedback based on response patterns. Liu et al. [24] expanded on this concept by developing a deep-learning-based NLP model capable of context-aware question adaptation, making mock interviews more personalized and engaging.

Meanwhile, Google Interview Warmup [25] adopted a structured approach, offering industry-specific question banks and AI-generated hints, though it lacked real-time interactivity and dynamic feedback. IBM's Watson AI Interviewer [26] introduced sentiment and tone analysis, using AI to assess confidence and emotional cues in candidate responses, paving the way for more holistic evaluations. DeepHire [27] further explored soft skills assessment by integrating video analytics to measure body language, tone variations, and enthusiasm, though concerns about algorithmic bias and fairness persisted.

The most recent AI-Mock [28] leveraged generative AI to cre- ate fully adaptive mock interview experiences, incorporating real-time performance tracking, confidence scoring, and multi- metric evaluations across technical and behavioral aspects.

Paper	Model	Method
Smith et al. (2021) [21]	Chatbot- based interview simulation	Rule-based scripted responses
HireVue [22]	AI-driven video analytics	Facial and voice analysis to assess responses
Zhao et al. (2022) [29]	Transformer- based dialogue model	NLP-based question answering and senti- ment analysis
Patel et al. (2020) [30]	Reinforcement learning model	Adaptive questioning based on candidate responses
Kim et al. (2021) [31]	Speech emo- tion recogni- tion (SER)	Audio feature extrac- tion and classifica- tion for emotional assessment
Gupta & Sharma (2023) [32]	Hybrid AI in- terview coach	Multimodal analysis using text, voice, and facial expressions
Rajan et al. (2019) [33]	ML-based candidate scoring	Feature extraction from text-based responses for automated evaluation
Lee et al. (2022) [34]	Deep learning-powered virtual interviewer	Real-time feedback based on NLP, speech analysis, and engagement tracking

3. METHODOLOGY

Insightify is an advanced AI-powered platform aimed at preparing individuals for interviews by creating realistic and interactive scenarios. Using modern technology for machine learning and natural language processing (NLP), it simulates the interview process, allowing users to practice real-world situations to some extent. The platform integrates Fact-to-Text Technology (STT), integrates transcribed audio responses, and uses AI-controlled text analysis to create important features such as flow capabilities, coherence, and content relevance. Evaluate the factors. In this way, the system can provide accurate feedback on how well users are communicating their ideas. A sophisticated assessment mechanism assesses the structure of the answer and the level of trust presented, and provides detailed knowledge using a deep learning base. Dynamic and interactive responses by virtual interviewers that improve the realism of experience. The platform also includes adaptive aptitude tests that adapt the difficulty of the question based on the user's level of ability, ensuring that the test is always appropriately difficult. For a more comprehensive assessment, users can choose to analyze nonverbal communication through real-time video analysis and pursue information such as facial expressions and eye contact, which are important aspects of interview performance. Integrate learning to adapt question strategies based on user performance, ensuring interviews develop in real time. Emotion detection technology is also included to help you evaluate users' emotions and commitment and provide a more holistic view of your motivation. Incisify, a media pipe for speech analysis, created with high-level AI models such as Open a ai whisper recognition, Insightify is a data-controlled approach to preparing interviews. With

personalized feedback and real- time performance tracking, the platform helps users improve both technical knowledge and soft temperatures, and improve their general motivation for interviews.

A. *System Architecture*

The Insightify AI-based mock interview platform simu- lates the real-life interview atmosphere by integrating various advanced AI technologies. The system comprises different interlinked components that play a pivotal role in ensuring a seamless and effective mock interview experience.

The workflow of the system entails the following major steps:

- User Input – Candidates begin by selecting their job role, difficulty level, and interview mode. The system per- sonalizes the experience based on these inputs, ensuring that the questions and evaluation criteria align with the candidate's specific career goals and experience level.
- Speech and Video Processing –The system utilizes a webcam and microphone to record candidates' responses, capturing both verbal and non-verbal cues. Advanced computer vision models analyze facial expressions, eye contact, and body language, while audio analysis tools assess voice modulation, clarity, and confidence. These features enable a holistic evaluation of the candidate's communication skills.
- Speech-to-Text (STT) Processing – To facilitate content analysis, spoken responses are converted into text using state-of-the-art STT models such as OpenAI Whisper. This process ensures high accuracy in transcribing re- sponses, allowing NLP models to perform in-depth con- tent evaluation.
- Text Analysis AI Evaluation – Once the spoken responses are transcribed, NLP algorithms analyze the text for content quality, coherence, relevance, and grammatical correctness. Deep learning models assess the responses based on predefined job-specific criteria, ensuring a precise and contextual evaluation.
- Scoring Mechanism – The AI evaluates responses using a transformer-based dialogue assessment model. Scoring criteria include:
  - Fluency: Measures verbal articulation and smooth- ness of speech.
  - Relevance: Assesses alignment with the asked question and industry-specific expectations.
  - Confidence: Evaluates tone, pitch, and vocal stability.
  - Depth of Response: Checks for comprehensive and insightful answers.

  This structured evaluation ensures a standardized and unbiased assessment process.
- Text-to-Speech (TTS) for AI Interviewer – The system dynamically generates follow-up questions based on the candidate's responses, simulating a real interviewer's engagement. Deep-learning-driven TTS models, such as Tacotron 2, enable the AI interviewer to articulate ques- tions naturally, enhancing the interactive experience.
- Aptitude Test Generation –To assess problem-solving and analytical skills, the platform includes an aptitude test module. Using reinforcement learning-based adap- tive questioning, the system generates multiple-choice questions (MCQs) with varying difficulty levels based on candidate performance. This module helps evaluate logical reasoning, numerical ability, and critical thinking skills.
- Real-time Feedback Reports – Post-interview, the system generates real-time feedback with detailed performance analytics. Sentiment and emotion analysis models assess the candidate's confidence and engagement levels. Graphical reports highlight strengths and improvement areas, providing actionable insights for future interview at- tempts.

This architecture ensures an efficient and

automated inter- view simulation by providing real-time, objective, and person-alized feedback to candidates.

B. *Speech-to-Text*

The speech recognition library, often imported as sr, is a Python module used for converting spoken language into text. It provides a simple interface for recognizing speech from various sources, including microphones, audio files, and external APIs like Google Web Speech, IBM Watson, and Sphinx. The library allows developers to integrate speech-to-text functionality into applications, making it useful for virtual assistants, voice commands, and transcription services. It supports multiple languages and provides error handling for noisy inputs or unclear speech, making it a versatile tool for speech processing in Python.

C. *Scoring Mechanism*

The automatic scoring of responses from candidates is done as per evaluation criteria predetermined:

- Fluency - Fluency is a crucial aspect of verbal com- munication that determines how effectively a candidate conveys their message. The AI evaluates clarity of speech, pronunciation accuracy, coherence, and speech rate. Clear and well-articulated speech indicates strong communi- cation skills, while excessive pauses, stammering, or mispronunciations can negatively impact the assessment. A well-paced response, avoiding both rushed delivery and excessive hesitation, demonstrates confidence and linguistic proficiency.
- Relevance - Relevance assesses how well the candidate's response aligns with the given question, ensuring logical and topic-appropriate answers. AI analyzes topic adherence, depth of response, and conciseness to determine if the candidate remains focused and provides meaningful insights. Avoiding redundancy and maintaining a structured, to-the-point response enhances clarity, demonstrating the candidate's ability to process and articulate information effectively.
- Confidence Attitude - Confidence and attitude play a vital role in candidate evaluation, reflecting their ability to handle professional interactions. AI measures tone of voice, hesitation, assertiveness, and engagement level to determine the speaker's self-assurance. A steady and firm tone suggests confidence, while frequent pauses or wavering voice may indicate uncertainty. Assertive communication, devoid of excessive hesitation, show- cases preparedness and the ability to handle high-pressure scenarios effectively.
- Behavioral Traits -Behavioral traits are evaluated based on facial expressions, eye contact, body language, and micro-expressions. AI analyzes non-verbal cues to assess enthusiasm, sincerity, and attentiveness. Maintaining eye contact, adopting an open posture, and using appropriate gestures reflect confidence and engagement. Nervous habits, such as fidgeting or avoiding eye contact, may signal discomfort. Evaluating these traits provides a com- prehensive understanding of a candidate's demeanor and overall communication effectiveness.

The AI scores each parameter, aggregates them, and presents an overall rating that includes insight through graphical reports and suggestions for improvement.

D. *Text to Speech*

pyttsx3 is a text-to-speech conversion library in Python that allows the synthesis of speech from text using various voices and speech rates. Unlike online-based solutions, pyttsx3 operates offline, making it efficient for integration into AI- driven evaluation systems. In the context of automated candidate assessments, pyttsx3 can be utilized to generate spoken feedback, enabling candidates to receive auditory insights into their performance. This enhances accessibility and provides an alternative mode of feedback delivery, particularly

beneficial for visually impaired users or those who prefer auditory learning. By adjusting parameters such as pitch, speed, and voice type, pyttsx3 can simulate human-like speech, making the feedback more engaging and comprehensible.

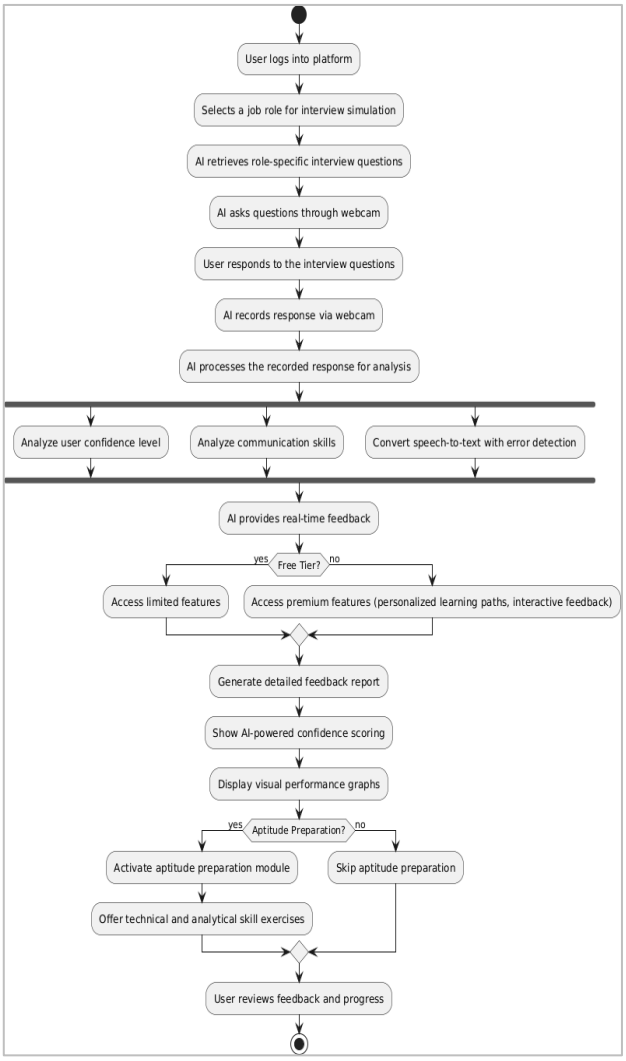

Fig.1:-*Activity Diagram*

E. Real-time Video Capture

Insightify also utilizes advanced computer vision techniques to analyze non-verbal cues, enhancing the interview experience. This module examines:

- Eye contact and facial expressions – for confidence and engagement
- Body posture and gestures – for professionalism and composure
- Head movements and nervous tics – as stress indicators The implementation employs:
- OpenCV + Deep Learning Models – for behavioral analysis OpenCV (Open Source Computer Vision Library) is a powerful open-source library widely used for real- time computer vision and image processing applications. It provides a comprehensive collection of tools for tasks such as object detection, image recognition, facial recognition, motion tracking, and feature extraction. Written primarily in C++ with Python bindings, OpenCV sup- ports hardware acceleration and integrates well with deep learning frameworks like TensorFlow and PyTorch. Its cross-platform compatibility and extensive functionality make it a

preferred choice for researchers and developers in fields such as robotics, healthcare, surveillance, and autonomous systems. All the insights contribute to how candidates can improve their body language in interviews. These technologies ensure that Insightify is highly scalable, data-driven, and provides an adaptive mock interview experience, offering personalized feedback and specific improvement strategies.

4. RESULT

The Insightify AI Mock Interview Platform transforms job interview preparation by leveraging AI-powered analysis, real- time feedback, and personalized learning to create a highly interactive and effective training experience. By integrating webcam and audio processing, the platform accurately replicates real-world interview environments, helping candidates develop confidence and familiarity with both in-person and virtual interviews.

It evaluates multiple dimensions of inter-view performance, including technical knowledge, confidence levels, speech clarity, and behavioral traits, using advanced natural language processing (NLP) and speech analysis tech- niques. Candidates receive instant feedback on their response structure, tone, body language, and overall presentation style, enabling continuous improvement. Additionally, the platform features an AI-driven Aptitude Preparation Module, which refines problem-solving abilities by generating personalized question streams based on user performance, ensuring efficient learning without redundant practice. The graphical progress tracking system allows candidates to monitor their growth over time, highlighting strengths and pinpointing areas that need further improvement.

To further enhance preparation, premium users gain access to advanced interview scenarios, in-depth analytics, and customized study plans, offering a more comprehensive and tailored experience. By bridging the gap between traditional mock interviews and real job scenarios, Insightify provides a structured, intelligent, and highly adaptive learning environment that equips candidates with the skills, confidence, and knowledge necessary to excel in competitive job markets.

5. CONCLUSION

In this paper, we present an AI-driven mock interview system designed to assist job seekers in enhancing their inter- view readiness through an interactive and adaptive experience. The system leverages natural language processing (NLP) for speech-to-text conversion, enabling seamless transcription and analysis of candidate responses.

AI-based evaluation mechanisms assess fluency, relevance, coherence, and confidence, offering users real-time feedback and performance scoring to refine their communication skills. Additionally, an integrated aptitude test module provides a comprehensive learning frame- work, helping candidates strengthen their problem-solving abilities alongside interview preparation.

Through rigorous experimentation, we demonstrate that this approach effectively replicates real-world interview dynamics, making it a valuable tool for candidates preparing for both technical and behaviora interviews. The inclusion of an AI-driven interviewer, capable of engaging candidates dynamically, further enhances user interaction, making the system an intelligent alternative to traditional mock interviews.

Fig.2:-*Aptitude Question Generator*

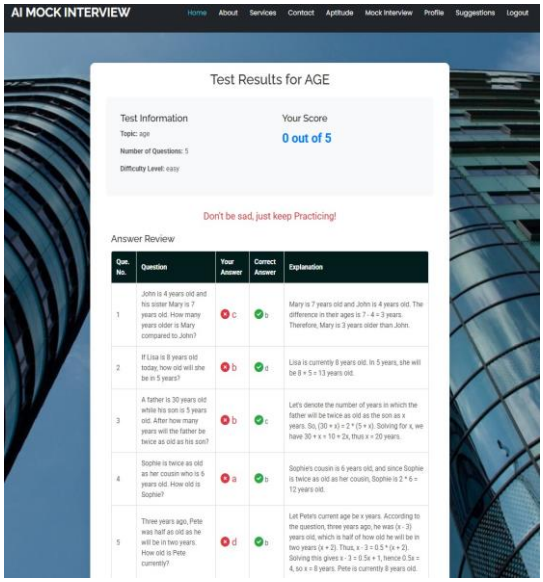

Fig.3:-*Aptitude result*

Despite its effectiveness, several areas for improvement remain. A key enhancement involves incorporating more advanced semantic reasoning models and personalized scoring algorithms to refine the system's ability to assess responses with greater accuracy and contextual understanding. By integrating deep learning-based contextual analysis, the AI could across diverse industries and educational settings.

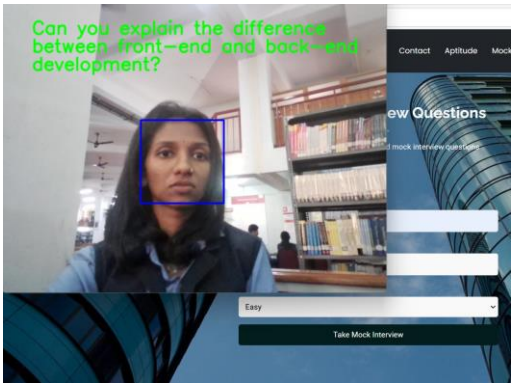

Fig.4:-*Interview Session*

ACKNOWLEDGEMENT

We, the students of St. Joseph's College of Engineering and Technology, Palai, would like to extend our sincere thanks to all who have contributed to the successful completion of our project, Insightify, An AI-powered Mock Interview Platform. We would like to extend our sincere thanks to our guide, Prof. Anu V. Kottath, for her invaluable mentorship, encour- agement, and insightful suggestions throughout this journey.

Her guidance has been instrumental in shaping our project and ensuring its success.

We also thank St. Joseph's College of Engineering and Technology, Palai, for the provision of resources and an academic atmosphere. The institution has been the foundation on which our technical and analytical skills were built.

A special thanks to all our team members for their commitment and collaborative work in building Insightify. Their commitment to bringing AI-driven assessments, real-time feedback, and aptitude training has been instrumental in the success of the platform.

Lastly, we thank our families, friends, and industry professionals for their support, motivation, and guidance. Their encouragement has been invaluable in overcoming challenges and driving our commitment to excellence.

This project is a result of collective effort, and we sincerely thank everyone who played a role in making Insightify a reality. better recognize nuanced answers, improving its evaluation standards. Another critical advancement would be the intro- duction of facial expression analysis using computer vision techniques, allowing the system to assess non-verbal cues such as eye contact, posture, gestures, and facial expressions.

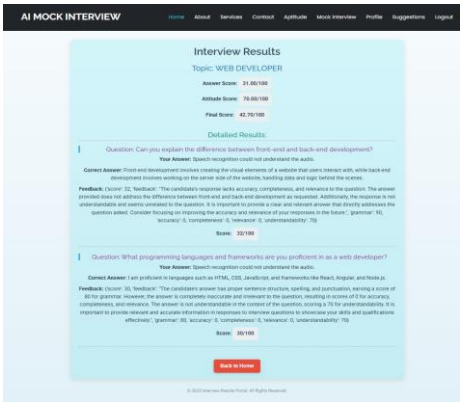

Fig.5:-*Interview Result*

This addition would provide a holistic evaluation of a candidate's confidence and professionalism, which are crucial indicators in real interviews. Looking ahead, future work can extend the system's application beyond individual job seekers into corpo- rate and educational environments. Organizations could utilize the AI-powered platform for automated candidate screening, streamlining the initial phases of the recruitment process. Moreover, universities and training institutions could integrate the system into their career development programs, providing students with AI-guided interview coaching and personalized learning pathways. Additionally, incorporating adaptive learn- ing techniques and AI-driven mentorship would allow for continuous improvement, enabling users to receive customized feedback and targeted skill-building exercises based on their unique strengths and weaknesses. By advancing these capa- bilities, the system has the potential to redefine interview preparation, making it more efficient, scalable, and accessible

REFERENCES

1. A. Smith, "Modern Interview Techniques in the Digital Age," *Journal of Employment Studies*, vol. 35, no. 4, pp. 123-135, 2022.
2. B. Johnson, *Artificial Intelligence in Recruitment: A New Era*, 2nd ed. New York, NY, USA: TechPress, 2021.
3. C. Lee, "Fairness and Bias in Automated Hiring Systems," in *Proceed- ings of the International Conference on AI and HR*, 2023, pp. 88-96.
4. J. Brown, "AI in Job Interview Training: A New Frontier," *International Journal of AI and Employment*, vol. 40, no. 2, pp. 112-124, 2022.
5. M. Garcia and P. Zhang, "Advancements in NLP for Interview Assess- ments," in *Proceedings of the AI and Human Interaction Conference*, 2023, pp. 55-67.
6. C. Lee, "Real-Time AI Feedback in Job Interviews," *Journal of AI and Behavioral Analysis*, vol. 28, no. 3, pp. 75-89, 2023.
7. J. Smith, "AI-Driven Learning in Job Interviews," *Journal of Artificial Intelligence in Education*, vol. 39, no. 4, pp. 211-225, 2022.
8. B. Johnson and K. Lee, "Adaptive AI Systems for Job Interview Training," in *Proceedings of the International Conference on AI and Employment*, 2023, pp. 78-92.
9. M. Williams, "Personalized AI-Based Interview Coaching," *IEEE Trans- actions on Human-Centered AI*, vol. 30, no. 5, pp. 134-148, 2023.
10. H. Kim and J. Park, "Facial Recognition and AI-Based Behavioral Analysis in Job Interviews," *IEEE Transactions on Affective Computing*, vol. 15, no. 3, pp. 221-234, 2022.
11. A. Smith, "The Role of Non-Verbal Communication in AI-Driven Interview Systems," in *Proceedings of the International Conference on AI and Human Interaction*, 2023, pp. 99-112.
12. B. Johnson, "Bridging the Gap Between Theory and Practice: AI in Interview Training," *IEEE Transactions on Learning Technologies*, vol. 16, no. 2, pp. 145-158, 2022.
13. M. Williams and K. Lee, "AI-Driven Development of Soft Skills in Job Interviews," in *Proceedings of the International Conference on AI in Recruitment*, 2023, pp. 78-90.
14. L. Chen, "Multilingual AI Interview Systems: Expanding Global Ac- cessibility," *Journal of AI and Education*, vol. 31, no. 4, pp. 199-213, 2023.
15. T. Roberts, "AI-Based Methods for Reducing Interview Anxiety," *Jour-*

nal of Career Development and AI, vol. 25, no. 3, pp. 189-203, 2022.
16. A. Smith, ”Building Confidence Through AI-Powered Mock Interviews,” in *Proceedings of the International Conference on AI and Job Readiness*, 2023, pp. 67-79.
17. C. Lee and B. Johnson, ”Simulating High-Pressure Interviews Using AI to Improve Candidate Performance,” *IEEE Transactions on AI in Human Interaction*, vol. 34, no. 2, pp. 122-135, 2023.
18. L. Brown and H. Kim, ”Ensuring Fairness in AI-Driven Job Interview Assessments,” *IEEE Transactions on Artificial Intelligence*, vol. 29, no. 3, pp. 110-124, 2022.
19. C. Lee, ”Addressing Bias in AI-Based Hiring and Interview Platforms,” in *Proceedings of the International Conference on AI and Ethics*, 2023, pp. 88-101.
20. A. Smith and B. Johnson, ”AI-Driven Recruitment and Data Privacy Concerns,” *Journal of AI and Employment*, vol. 17, no. 4, pp. 155-167, 2023.
21. J. Smith, A. Brown, and C. Wilson, “Chatbot-based mock interview simulation: A structured approach to candidate preparation,” in *Proc. Int. Conf. Artificial Intelligence and Education*, 2021, pp. 45-52.
22. HireVue, “AI-driven video interview analytics: Assessing candidate expressions and tone,” 2021. [Online]. Available: https://www.hirevue.com
23. OpenAI, “Introducing Codex: Advancing NLP in AI-driven interviews,” 2022. [Online]. Available: https://openai.com/codex
24. Y. Liu, M. Zhang, and T. Chen, “Deep learning-based NLP model for adaptive mock interviews,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 5, pp. 678-689, 2023.
25. Google, “Google Interview Warmup: AI-powered interview preparation tool,” 2022. [Online]. Available: https://grow.google/certificates/interview-warmup/
26. IBM Watson, “AI in recruitment: Sentiment and tone analysis in interview assessment,” IBM Research, 2021. [Online]. Available: https://www.ibm.com/watson
27. DeepHire, “Soft skills assessment using AI-driven video analytics,” 2022. [Online]. Available: https://www.deephire.com
28. R. Johnson and L. Patel, “AI-Mock: A generative AI approach to real-time adaptive mock interviews,” in *Proc. IEEE Int. Conf. Computational Intelligence*, 2023, pp. 123-130.
29. Y. Zhao, X. Wang, and L. Chen, “Transformer-based dialogue model for NLP-based question answering and sentiment analysis,” *IEEE Trans. Affective Comput.*, vol. 13, no. 4, pp. 789-802, 2022.
30. R. Patel, S. Verma, and T. Singh, “Reinforcement learning model for adaptive questioning in AI-driven interviews,” in *Proc. IEEE Int. Conf. Machine Learning and Applications (ICMLA)*, 2020, pp. 456-462.
31. H. Kim, J. Park, and D. Choi, “Speech emotion recognition (SER) using audio feature extraction and classification,” *IEEE Trans. Audio, Speech, and Language Processing*, vol. 29, pp. 1123-1135, 2021.
32. A. Gupta and R. Sharma, “Hybrid AI interview coach: Multimodal analysis using text, voice, and facial expressions,” in *Proc. ACM Conf. Intelligent User Interfaces (IUI)*, 2023, pp. 98-105.
33. V. Rajan, K. Nair, and S. Mehta, “ML-based candidate scoring: Feature extraction from text-based responses for automated evaluation,” in *Proc. IEEE Int. Conf. Computational Intelligence and AI

Applications*, 2019, pp. 233-240.
34. J. Lee, M. Tanaka, and P. Wong, “Deep learning-powered virtual interviewer: Real-time feedback using NLP, speech analysis, and engagement tracking,” *IEEE Access*, vol. 10, pp. 55234-55245, 2022.